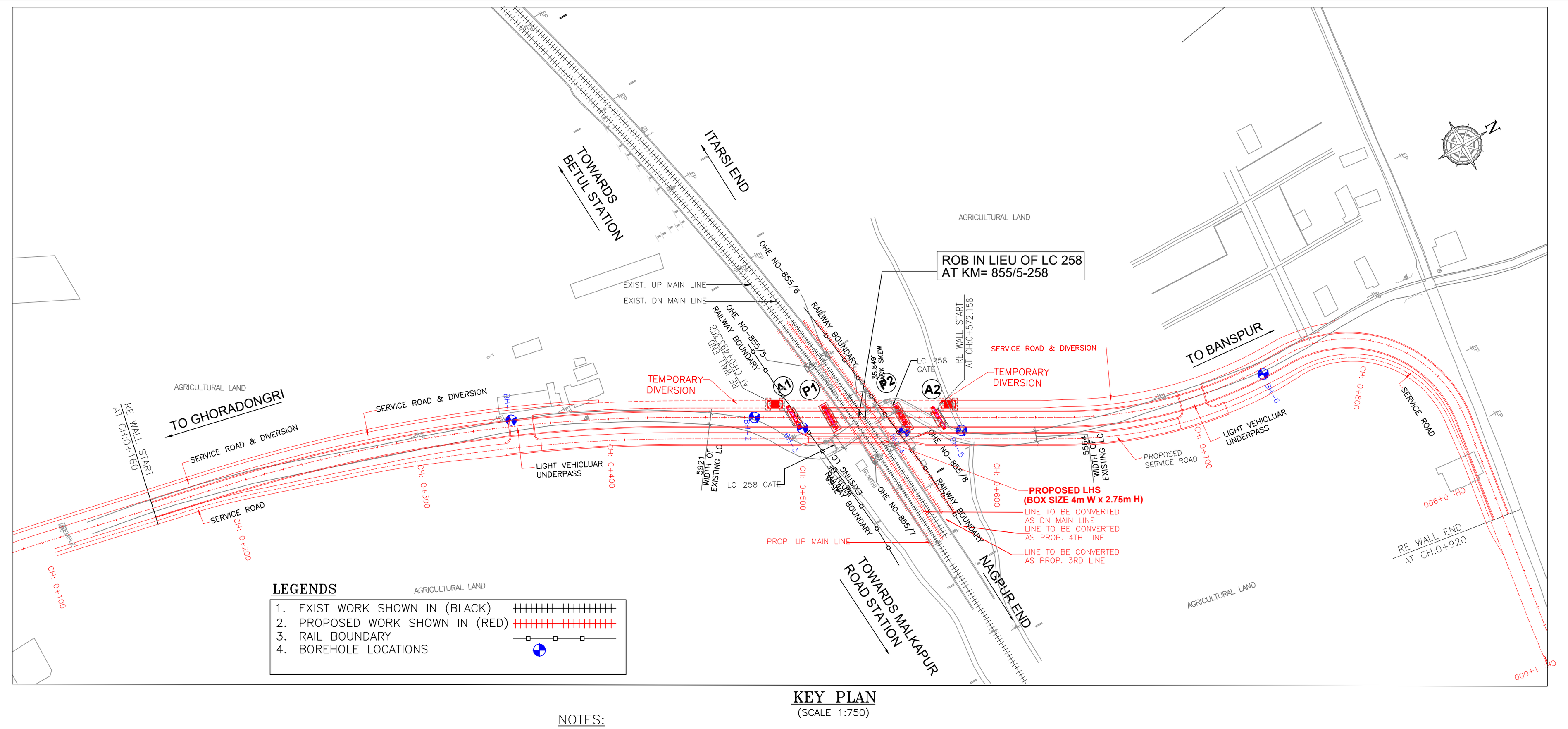

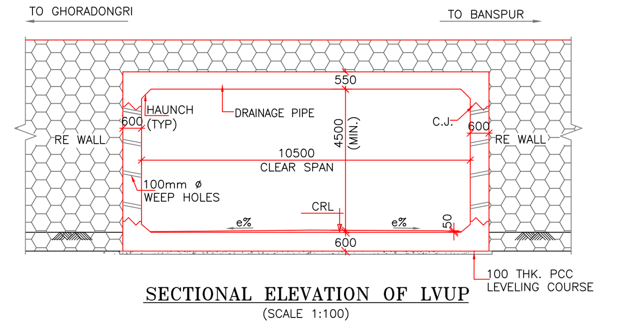

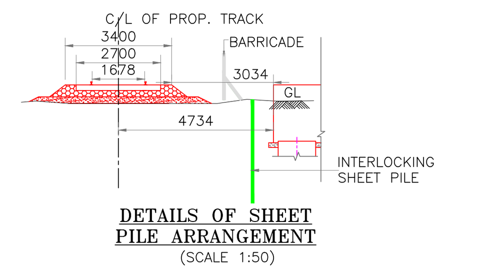

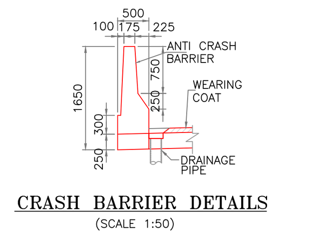

NOTES:

1. ALL DIMENSIONS ARE SHOWN IN MM, LEVELS ARE IN METER & CHAINAGE ARE IN KILOMETERS UNLESS OTHERWISE MENTIONED.
2. NO DIMENSIONS SHALL BE SCALED FROM DRAWING. ONLY WRITTEN DIMENSIONS TO BE FOLLOWED.
3. ALL THE DIMENSIONS SHOWN SHOULD BE VERIFIED AT SITE BEFORE TAKING UP DESIGN WORK IN HAND. ANY DISCREPANCY NOTED SHOULD BE FORWARDED THROUGH ALTERATION DULY SIGNED.
4. SUITABLE SPEED RESTRICTION, IF REQUIRED SHOULD BE IMPOSED BEFORE TAKING UP THE EXECUTION OF WORK WITH THE APPROVAL OF DIVISIONAL ENGINEER IN–CHARGE.
5. DURING CONSTRUCTION OF ROB ALL THE EXISTING STRUCTURES/TRACKS SHOULD BE PROTECTED BY PROVIDING PROPER SHORING ARRANGEMENT IF REQUIRED WITH THE APPROVAL OF DIVISIONAL ENGINEER IN CHARGE.
6. REPRESENTATIVE OF SIGNALING & TELECOM DEPARTMENT SHOULD BE AVAILABLE AT SITE DURING EXECUTION WITHIN VICINITY OF SIGNALING/TELECOM CABLES.
7. ALL THE PRE–FABRICATED STEEL GIRDERS SHOULD BE LAUNCHED WITH SUITABLE LAUNCHING SCHEME AS PER SITE REQUIREMENT, WITH THE APPROVAL OF RAILWAY AND AFTER OBTAINING CRS SANCTION.
8. TEMPORARY LC SHALL BE CLOSED SIMULTANEOUSLY WITH THE COMMISSIONING OF ROB.
9. TVU OF LC IS 37368 OF YEAR MARCH 2022 (ENGG. INTERLOCKED).
10. FOR INSPECTION OF RAILWAY SPAN, PROVISION OF INSPECTION CRADLE FOR BOTH SIDE OF PIER/ABUTMENT WITH RAILING AND LADDER SHOULD BE MADE.
11. THE APPROVAL OF GAD BY CBE OFFICE IS ONLY FOR SPAN CONFIGURATION, TYPES OF SUPERSTRUCTURE, SUBSTRUCTURE & FOUNDATION. THE SIZE AND DIMENSION OF SUBSTRUCTURES AND FOUNDATIONS ARE INDICATIVE ONLY AND IT SHOULD BE DESIGNED, OPTIMIZED AND APPROVED BY COMPETENT AUTHORITY ON CAO(C) OFFICE.
12. DEPTH OF FOUNDATION MAY BE ALTERED BY ENGINEER IN–CHARGE TO SUIT FOUNDING STRATA MET AT SITE AS PER BEARING CAPACITY.
13. NO PLASTERING AND CEMENT WORK PROTECTIVE COATING OVER THE CONCRETE IS PERMITTED. ANY BAD QUALITY WORK WILL BE SUMMARILY REJECTED.
14. THE VISIBILITY OF THE SIGNAL SHALL NOT BE OBSTRUCTED.
15. WEEP HOLES AT 1000MM C/C BOTH HORIZONTALLY AND VERTICALLY STAGGERED SHALL BE PROVIDED FOR APPROACH SECTION.
16. SAFETY, RELIABILITY AND LONG TERM DURABILITY OF THE OVERHEAD STRUCTURE SHALL BE ENSURED DURING THE DESIGN AND EXECUTION STAGES AS PER CENTRAL RAILWAY LETTER NO. W.641 BR.POLICY DATED 10.10.2018.
17. WHILE SUBMITTING THE DESIGN OF ROB FOR APPROVAL TO HEAD QUARTERS, GUIDELINES ISSUED BY HD.QRS. OFFICE VIDE LETTER NO. W.294BR.PROOF CHECKING DATED 27.08.2018 RAILWAY CROSSING STRUCTURE SHALL BE CARRIED OUT BY THE OWNER.
18. UTILITIES IN RAILWAY PORTION AND APPROACHES ARE REQUIRED TO BE SHIFTED DURING EXECUTION.
19. SITE OFFICE AT THE BRIDGE SITE OR AT NEARBY LOCATION DECIDED BY THE RAILWAY SHALL BE PROVIDED AND SHALL BE MAINTAINED DURING THE PERIOD OF CONSTRUCTION OF ROB.
20. VERTICAL CLEARANCE FROM HIGHEST RAIL LEVEL TO BOTTOM LEVEL OF GIRDER SHALL NOT BE LESS THAN 6150mm.
21. EXECUTION OF ANY WORK WITHIN RAILWAY BOUNDARY FOR CONSTRUCTION OF ROB SHALL BE DONE ONLY IN PRESENCE OF RAILWAY'S AUTHORIZED REPRESENTATIVE TO ENSURE THE SAFETY OF RUNNING TRAINS. NO HEAVY MACHINERY WILL WORK ALONG THE TRACK WITHOUT THE PRESENCE OF RAILWAY AUTHORITY.
22. SIZES OF STRUCTURAL MEMBERS ARE TENTATIVE AND MAY BE REVISED AT THE DETAILED DESIGN STAGE.
23. XPM PROTECTION FRAMES SHALL BE PROVIDED FOR ELECTRIFIED SECTION AND FIXED WITH PARAPET OF ROB ON EACH LINE ON BOTH SIDES OF ROB.
24. GALVANIZED SACRIFICIAL SHUTTERING TO BE USED FOR CASTING OF DECK SLAB.
25. SHAPES AND SIZE OF STEEL PLATE GIRDER. POT–PTFE/SPHERICAL BEARINGS, PEDESTAL, PIER/ABUTMENT, PIER/ABUTMENT CAP AND FOUNDATION SHOWN IN THIS DRAWING ARE TENTATIVE AND ARE SUBJECTED TO CHANGE IN FINAL EXECUTION AFTER APPROVAL OF DESIGN BY RAILWAY.
26. DESIGN SPEED 50KMPH OVER ROB.
27. A SEPARATE AGREEMENT BETWEEN RLY & ROAD AUTHORITY WILL BE EXECUTED FOR FUTURE MAINTENANCE OF ROB & SURFACE.
28. COMPLETE ROB IS BEING CONSTRUCTED BY RAILWAY AT OWN COST.
29. AFTER COMPLETION RAILWAYS LIABILITY IS TO MAINTAIN RAILWAY PORTION SPAN i.e.FROM A1–A2.
30. AFTER COMPLETION PWD WILL MAINTAIN APPROACH PORTION SPAN i.e. RE WALL APPROACHES.
31. AREA OF LAND ACQUISITION REQUIRED FOR THE PROPOSED SCHEME IS 11020 sqm. LAND ACQUISITION WILL BE CARRIED OUT BY RAILWAY.
32. SUITABLE UTILITY DUCT SHALL BE PROVIDED BELOW THE FOOTPATH AS PER INSTRUCTION OF ENGINEER IN CHARGE.
33. FABRICATION SHALL BE DONE BY RDSO APPROVED FIRM AND INSPECTION SHALL BE DONE IN ACCORDANCE WITH THE RAILWAY BOARD'S LETTER NO. 2017/16/CE–III/BR/GIRDER DATED 26.05.2018
34. WORK SHALL BE CARRIED OUT IN THE FOLLOWING PHASES:
    34.1. TEMPORARY DIVERSION SHALL BE EXECUTED AND LIFTING BARRIERS SHALL BE SHIFTED TO CREATE TEMPORARY LC.
    34.2. EXISTING LC SHALL BE CLOSED.
    34.3. ROB SHALL BE CONSTRUCTED ALONG WITH SERVICE ROAD CONNECTIONS AND COMMISSIONED.
    34.4. TEMPORARY LC SHALL BE CLOSED.
    34.5. LHS SHALL BE CONSTRUCTED.
35. GAD IS PREPARED AS PER PROJECT SHEET NO. C. RLY HQ (E) PLAN NO. GM (E) CSMT – 31638 / 12 OF 14 AND APPROVED LAND PLAN SHEET NO. BARSALI–DHARAKHOH No. 7/31.
36. STANDARD CRASH BARRIER AS PER IRC–5 2015 AND IRC–119 2015 OF MIN. 30m LENGTH IN BOTH THE APPROACHES OF ROB SHALL BE PROVIDED.
37. LC WILL BE CLOSED AFTER CONSTRUCTION OF ROB AFTER OBTAINING NOC FROM DISTRICT COLLECTOR'S OFFICE.
38. WORK SHALL BE STARTED AFTER OBTAINING NOC FROM DISTRICT COLLECTOR'S OFFICE.
39. SEPARATE GAD FOR LHS WILL BE PROCESSED & APPROVED BY COMPETENT AUTHORITY.
40. FOUNDATION & STRUCTURAL DETAILS ARE TENTATIVE AND SHALL BE PROVIDED AS REQUIRED FROM GEOTECHNICAL & STRUCTURAL DESIGN. HOWEVER, CLEARANCES INDICATED SHALL NOT BE REDUCED.
41. THE NUMBER OF GIRDERS REDUCED TO 5 FROM THE REFERRED RDSO STANDARD DRAWING

TECHNICAL NOTES

1. GUARD RAIL TO BE PROVIDED AS PER 'RAILWAY STANDARD DRAWING'.
2. 36.00m SPAN COMPOSITE GIRDER DWG NO:– RDSO/B–11775/R TO 11775/12R IS PROVIDED FOR RAILWAY SPAN.
3. FOR FABRICATION OF STEEL GIRDERS, FOLLOW RDSO GUIDELINES BS–110, REVISED 112 AND SPECIFICATION OF B–1.
4. SAND BLASTING OF STEEL GIRDERS/MEMBERS METALIZING AND PROTECTIVE COATS OF APPROVED QUALITY PAINTS AS PER IRBM 1998 OF PARA 218.
5. PROOF LOAD TEST AS PER IRC SP 51 SHOULD BE CONDUCTED AFTER COMMISSIONING OF ROB.
6. POT–PTFE/SPHERICAL BEARING IS ADOPTED WITH 30 YEAR GUARANTEE CERTIFICATE.
7. REINFORCEMENT BARS SHALL BE H.Y.S.D OF GRADE Fe 500D CONFORMING TO IS 1786.
8. STRIP SEAL TYPE EXPANSION JOINTS SHALL BE PROVIDED AS PER LATEST MOST REQUIREMENT. CONTRACTOR SHOULD FURNISH DESIGN/DRAWING OF EXPANSION JOINT FROM FIRMS APPROVED BY MOST AT VARIOUS RELEVANT AMBIENT TEMPERATURES TO SUIT SITE CONDITIONS WITH THE APPROVAL OF ENGINEER IN CHIEF.
9. DESIGN & DRAWINGS SHALL BE APPROVED BY IIT/NIT OR APPROVED INSTITUTION.

REFERENCES

1. 36.0m COMPOSITE GIRDER: RDSO/B–11775/R1 to RDSO/B–11775/R1/7
2. 18.0m COMPOSITE GIRDER: RDSO/B–11772/R1 to RDSO/B–11772/R1/7

DIVISION APPROVAL		HQ APPROVAL	
DESIGNATION	SIGNATURE	DESIGNATION	SIGNATURE
DRM-NGP	MANISH AGARWAL Digitally signed by MANISH AGARWAL Date: 2024.03.26 15:44:05+05'30'	CBE	Digitally signed by SUNIL KUMAR GUPTA Date: 2024.12.04 13:06:21 +05'30'
CPM-GSU-NGP	NAVIN NISCHAY PATIL Digitally signed by NAVIN NISHCHAY PATIL Date: 2024.03.26 17:43:11 +05'30'	DY.CE-BR-1	Digitally signed by ROHIT YADAVRAO THAWRE Date: 2024.12.04 13:00:39 +05'30'
SR.DOM-NGP	KRISHNATH SHAMRAO PATIL Digitally signed by KRISHNATH SHAMRAO PATIL Date: 2024.03.22 10:24:12 +05'30'	SSE-ROB-RUB-HQ	Digitally signed by KULDEEP NIMESH Date: 2024.12.04 12:55:11 +05'30'
SR.DEE-TRD-NGP	MOHAN CHOUDHARY Digitally signed by MOHAN CHOUDHARY Date: 2024.03.22 11:21:41 +05'30'		
SR.DSTE-CO-NGP	DHIRAJ KUMAR DWIVEDI Digitally signed by DHIRAJ KUMAR DWIVEDI Date: 2024.03.22 11:52:41 +05'30'		
SR.DEN-CO-NGP	RAJESH CHIKHALE Digitally signed by RAJESH CHIKHALE Date: 2024.03.21 15:26:01 +05'30'		
SR.DEN-N-NGP	SACHIN WAMANRAO GANER Digitally signed by SACHIN WAMANRAO GANER Date: 2024.03.21 15:05:16 +05'30'		
SR.DEN-GSU-NGP	MUKESH KUMAR SIROLIYA Digitally signed by MUKESH KUMAR SIROLIYA Date: 2024.03.23 15:02:41 +05'30'		
SSE - BR - DRG-NGP	SACHIN BABURAO PATIL Digitally signed by SACHIN BABURAO PATIL Date: 2024.03.21 12:42:23 +05'30'		
DYCE-C-BZU	SUJIT KUMAR SINGH Digitally signed by SUJIT KUMAR SINGH Date: 2024.06.26 13:09:14 +05'30'		

LOADING STANDARD	IRC-6 2017 SEISMIC ZONE - III
CENTRAL RAILWAY	
GENERAL ARRANGMENT DRAWING	
DIVISION	NAGPUR
SECTION	ITARSI-AMLA (ET-AMF)
EXECUTIVE AGENCY	GATI SHAKTI UNIT, NAGPUR DIVISION
CONSULTANT (IF ANY)	CEG CONSULTING ENGINEERS GROUP
PROJECT (PROJECT ID)	CONSULTANCY SERVICES FOR DETAILED DESIGN DRAWINGS OF 7 (SEVEN) ROAD OVER BRIDGES IN NAGPUR DIVISION OF CENTRAL RAILWAY. (UID NO.- 01.03.30.21.1.57.038 - RAILWAY COST)
DRAWING TITLE	**CONSTRUCTION OF ROB WITH LHS IN LIEU OF LC-258 AT RAILWAY KM: 855/5-7 BETWEEN BETUL - MALKAPUR STATIONS ON ITARSI - AMLA SECTION (ON GHORADONGRI - BANSPUR ROAD) SPAN: 1x18m C.G. + 1x36m C.G. + 1x18m C.G.**
DRAWING NO.	DIVISION: GSU–NGP–CR–R–041 CONSTRUCTION: ROB–D244–NGP–LC 258–DRG–100 HEAD QUARTER: GM(W)BB/P-13688-ROB
DATE:	26-11-2024
SCALE:	SCALE =1:100 / (AS SHOWN)